

Fluid Mechanics II (ME 221) – Midsemester Exam (Maximum Marks: 60)
 Department of Mechanical Engineering, Indian Institute of Technology Guwahati

- Timing is Sunday, 26 February, 2023, 09-11 AM.
- Make suitable assumptions wherever necessary and clearly mention them in the solution.
- Getting final answers correct without showing intermediate steps will not be accepted and no marks will be awarded.
- Wherever required, it is necessary to show the control volume/coordinates axis/origin etc.
- Take $g = 9.81 \text{ m/s}^2$

1. (10 Marks) A long plate drags an oil (density = 872.5 Kg/m^3 and viscosity = 0.155 N-s/m^2) film as it moves upwards (see figure 1). After a long time, the film thickness h is found to be 1.04 cm . Find the velocity distribution inside the film when there is no net flow of oil.

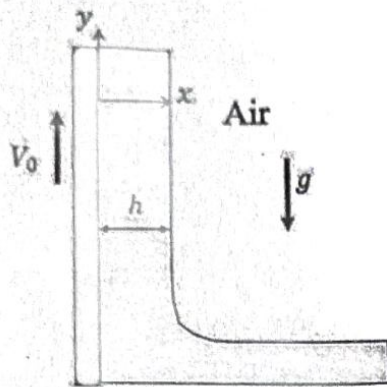


Figure 1

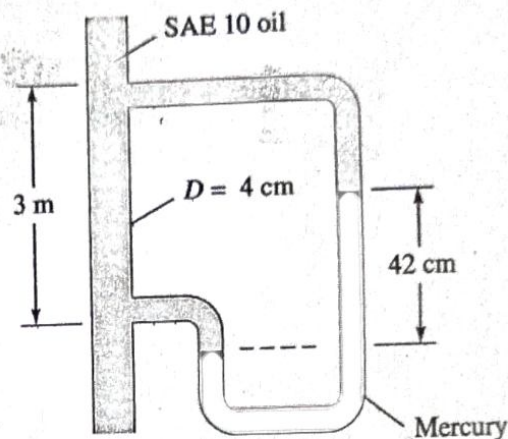


Figure 2

2. (6 Marks) SAE 10 oil (density = 870 Kg/m^3 and viscosity = 0.104 Kg/m-s) flows through the 4-cm-diameter vertical pipe. For the mercury (density = 13550 Kg/m^3) manometer reading $h = 42 \text{ cm}$ shown in figure 2 (assuming laminar flow through the pipe)
- State the direction of flow. [2 Marks]
 - Calculate the volume flow rate in m^3/h . [4 Marks]
3. (12 Marks) Consider two-dimensional laminar boundary-layer flow along a flat plate. Assume that the velocity profile in the boundary layer is sinusoidal, $u/U_\infty = \sin(\alpha y/\delta + b)$
- Find the value of constants α and b . [2 Marks]
 - Find the boundary layer thickness $\delta(x)$ in terms of Re_x . [3+1+2 Marks]
 - Find the displacement thickness $\delta^*(x)$ in terms of Re_x . [2 Marks]
 - Find the total frictional force per unit width on the flat plate of length L . [2 Marks]
4. (7 Marks) A spherical ball of diameter $D = 5 \text{ cm}$ and specific gravity $SG = 0.5$ is dropped into water ($\rho = 998 \text{ kg/m}^3$) at inlet velocity $V_0 = 10 \text{ m/s}$. It will travel a depth h and then pop out again. Assuming a constant drag coefficient $C_D = 0.47$ and acceleration due to gravity $g = 9.81 \text{ m/s}^2$, find h .
5. (13 Marks) Consider a steady flow of an incompressible Newtonian fluid through a parallel plate channel. The height of the channel (H) is very small as compared to its other two dimensions. The flow can be treated as the parallel flow in the axial direction. The fluid properties are assumed to be constant. The flow is driven by the upper plate only, which is moving in its own plane under the

influence of a constant forcing F . Some kind of pre-coated chemical layer at the lower wall alters the flow velocity in the field by enforcing a new condition at fluid-solid interface and is given as: $u = \lambda \frac{\partial u}{\partial n}$, where u is the fluid velocity tangent to the surface and n is the normal vector pointing

towards the fluid. The parameter λ is known as 'slip length'.

- a) Derive the pertinent equation(s) that govern the flow dynamics for this problem. [3 Marks]
 - b) By solving the governing transport equation(s) obtain the velocity distribution in the channel. [4 Marks]
 - c) Write the boundary conditions that you have considered for obtaining the velocity distribution. [3 Marks]
 - d) Also, clearly state the assumptions you have considered for this analysis. [2 Marks]
 - e) What would be the velocity profile if compared with that of the profile for $\lambda = 0$. [1 Marks]
6. (4 Marks) A radial flow pump is used to supply water to IIT Guwahati campus through a circular pipe having length 300 m. Based on the average flow velocity of water, the Reynolds number is calculated as 2000. Up to an initial 50 m from the pump outlet, no support is needed for the pipe, while institute engineer needs to calculate the wall shear stress for designing rest of the pipe line. The engineer is reluctant to perform involved mathematical calculations; instead he wants to get an expression for the wall shear stress, which can be related to the directly measurable parameter(s). Clearly writing the justification(s)/steps, derive an expression of the wall shear stress. [2+2 Marks]
7. (8 Marks) Water of kinematic viscosity, $\nu = 1.04 \times 10^{-6} \text{ m}^2/\text{s}$ is flowing steadily over a smooth flat plate at zero angle of attack with a velocity 1.6 m/s. The length of the plate is 0.3 m. Assume a cubic velocity profile calculate the following:
- a) The thickness of boundary layer at 15 cm from the leading edge. [3 Marks]
 - b) The rate of growth of boundary layer at 15 cm from the leading edge. [2 Marks]
 - c) The total drag coefficient on one side of the plate. [3 Marks]